# DAVID CALEB CHAPARRO OROZCO

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## PROFESSIONAL SUMMARY

Senior Data Scientist / ML Engineer with 5+ years of experience deploying end-to-end AI solutions in production: Computer Vision, NLP, fraud detection, predictive analytics, Data Engineering and MLOps. Delivered real-time facial verification (86–90% precision, ~50 ms/image), predictive maintenance (~30% less downtime) and ML-based segmentation (+25% campaign effectiveness) on AWS, Azure, Google Cloud, IBM Cloud and Snowflake.

## TECHNICAL SKILLS

**Core:** Python, SQL, Machine Learning, MLOps, Data Engineering, ETL/ELT, Anomaly Detection, Bash
**ML / AI:** PyTorch, TensorFlow, Keras, Scikit-learn, BERT/RoBERTa, InsightFace, OpenCV, LSTM, Prophet, Clustering (K-Means, DBSCAN, HDBSCAN, GMM), Reinforcement Learning, RAG / LLM APIs
**Data & MLOps:** Airflow, Prefect, dbt, Spark, Databricks, FastAPI, Docker, Pandas, NumPy, Data Quality, RBAC, Power BI
**Cloud & Databases:** AWS (Lambda, S3, EC2), Azure Data Factory, Google BigQuery / Cloud Storage, IBM Cloud, Snowflake, NoSQL

## PROFESSIONAL EXPERIENCE

**Freelance Data Scientist | Multiple Clients** 12.2025 – Present

- Developed clustering solutions (K-Means, DBSCAN, HDBSCAN, GMM, hierarchical) with Scikit-learn for healthcare, banking, customer, product, financial and geospatial segmentation.
- Applied unsupervised learning to clinical data from a U.S. hospital to identify patient profiles and support early chronic kidney disease detection.
- Segmented industrial IoT sensor patterns with K-Means/DBSCAN as preprocessing for LSTM and Prophet forecasting models.

**Data Scientist | APIUX (Apiux Tecnologia SpA) | Client: National Registry of Colombia** 07.2025 – 12.2025

- Designed an AI inference pipeline on Snowflake and Airflow for real-time facial biometric verification (512-dim embeddings, ~50 ms/image, 86–90% precision).
- Built a Snowflake-backed fraud detection platform with FastAPI microservices for multi-criteria risk scoring, including RBAC, audit logging and data-quality validation.
- Automated large-scale ETL workflows in Airflow for ingestion, transformation, monitoring and fraud-risk reporting.

**Data Scientist | Freelancer | Multiple Clients** 12.2023 – 07.2025

- Built predictive maintenance solutions for industrial IoT (LSTM/Prophet, anomaly detection), contributing to ~30% lower equipment downtime.
- Developed ML-based customer segmentation (K-Means/DBSCAN) that improved targeted campaign effectiveness by 25%.
- Built multilingual NLP pipelines (BERT/RoBERTa) for real-time sentiment analysis and a RAG-based document retrieval and classification workflow using Claude and OpenAI APIs.
- Built dynamic pricing models using regression and reinforcement learning with SQL/NoSQL databases and external APIs.
- Architected ELT/ETL pipelines with Snowflake/dbt, Azure Data Factory, BigQuery, Databricks, Prefect and Airflow, cutting manual intervention by ~40%; delivered Power BI dashboards and Bash automation.

**Cloud Data Engineer | Tata Consultancy Services (TCS)** 05.2023 – 11.2023

- Built serverless ETL pipelines with Python and AWS Lambda, S3 and EC2, delivering reliable datasets for analytics and supporting 97% customer satisfaction and SLA adherence.
- Implemented cloud data governance with IAM-based access control, column-level encryption and observability dashboards.

**Data Engineer | Bancolombia (via SETI / IGERENCIA)** 01.2022 – 10.2022

- Built Python microservices and SQL data models for automated ETL workflows; deployed AWS Lambda functions for ML inference and cloud integrations.
- Developed AI projects on Azure and IBM Cloud for real-time decision-making in financial applications.
- Optimized databases and stored procedures to improve performance and response times; established data-quality standards that reduced inconsistencies across reporting systems.

**Data Engineer | Institución Universitaria de Envigado** 06.2020 – 12.2020

- Optimized databases and enforced data-quality standards across institutional AI projects, improving dataset robustness for analysis and model development.

## EDUCATION

**B.Sc. in Software Engineering**, Institución Universitaria de Envigado 2019 – 2024

## CERTIFICATIONS

Python for Data Scientists (LinkedIn) · Machine Learning & Advanced Analytics (SAPIENCIA) · CCNA Enterprise Networking (Cisco) · Introduction to IoT (Cisco)

## LANGUAGES

Spanish (Native) · English (B2)